

The theory of rational choice

The theory of rational choice is that a decision-maker chooses the best action according to her preferences, among all the actions available to her. The action chosen by a decision-maker is at least as good, according to her preferences, as every other available action.

The theory is based on a model with two components: a set A consisting of all the **actions** that, under some circumstances, are available to the decision-maker, and a specification of the decision-maker's **preferences**. In any given situation the decision-maker is faced with a subset of A , from which she must choose a single element. The decision-maker knows this subset of available choices, and takes it as given; in particular, the subset of actions is not influenced by the decision-maker's preferences.

As to preferences, we assume that the decision-maker, when presented with any pair of actions, knows which of the pair she prefers, or knows that she regards both actions as equally desirable. Preferences can be seen as a metric.

Instead of specifying preferences for each possible pair of actions, we can “represent” the preferences by a **payoff function** $u()$ (aka utility function), which associates a number with each action in such a way that actions with higher numbers are preferred. For any actions a in A and b in A , $u(a) > u(b)$ if and only if the decision-maker prefers a to b .

EXAMPLE 5.2 (Payoff function representing preferences) A person is faced with the choice of three vacation packages, to Havana, Paris, and Venice. She prefers the package to Havana to the other two, which she regards as equivalent. Her preferences between the three packages are represented by any payoff function that assigns the same number to both Paris and Venice and a higher number to Havana. For example, we can set $u(\text{Havana}) = 1$ and $u(\text{Paris}) = u(\text{Venice}) = 0$.

A decision-maker's preferences, in the sense used here, **convey only ordinal information**. They may tell us that the decision-maker prefers the action a to the action b to the action c , for example, but they do not tell us “how much” she prefers a to b , or whether she prefers a to b “more” than she prefers b to c .

It may be tempting to think that the payoff numbers attached to actions by a payoff function convey intensity of preference—that if, for example, a decision-maker's preferences are represented by a payoff function u for which $u(a) = 0$, $u(b) = 1$, and $u(c) = 100$, then the decision-maker likes c a lot more than b but finds little difference between a and b . But a payoff function contains no such information! The only conclusion we can draw from the fact that $u(a) = 0$, $u(b) = 1$, and $u(c) = 100$ is that the decision-maker prefers c to b to a .

if we assume that a decision-maker who is indifferent between two actions sometimes chooses one action and sometimes the other, not every collection of choices for different sets of available actions is consistent with the theory. Suppose, for example, we observe that a decision-maker chooses a whenever she faces the set $\{a, b\}$, but sometimes chooses b when facing the set $\{a, b, c\}$.

The fact that she always chooses a when faced with $\{a, b\}$ means that she prefers a to b (if she were indifferent then she would sometimes choose b). But then when she faces the set $\{a, b, c\}$ she must choose either a or c , never b . Thus, her choices are inconsistent with the theory.

EXAMPLE: if you choose the same dish from the menu of your favorite lunch spot whenever there are no specials then, regardless of your preferences, it is inconsistent for you to choose some other item from the menu on a day when there is an off-menu special.

Strategic games

A strategic game is a model of interacting decision-makers. In recognition of the interaction, we refer to the decision-makers as **players**. Each player has a set of possible **actions**. The model captures interaction between the players by allowing each player to be affected by the actions of all players, not only her own action. Specifically, each player has preferences about an **action profile** (i.e. a combination of choices of player' choices).

DEFINITION 11.1 (Strategic game with ordinal preferences) A strategic game (with ordinal preferences) consists of

- a set of players
- for each player, a set of actions
- for each player, preferences over the set of action profiles.

EXAMPLE: two players play even/odd game.

- Players: player1 (even), player2 (odd)
- Actions: say an odd number, say an even number)
- Action profile (even-even, even-odd, odd-even, odd-odd).
 - Preferences of player1 (even-even, odd-odd, even-odd, odd-even)
 - Preferences of player2 (even-odd, odd-even even-even, odd-odd)

As in the model of rational choice by a single decision-maker, it is convenient to specify the players' preferences by giving payoff functions that represent them. **These payoffs have only ordinal significance.**

- Time is absent from the model.
- The players choose their actions "simultaneously" in the sense that no player is informed, when she chooses her action, of the action chosen by any other player.

Prisoner's Dilemma

Two suspects in a major crime are held in separate cells. There is enough evidence to convict each of them of a minor offense, but not enough evidence to convict either of them of the major crime unless one of them acts as an informer against the other (finks). If they both stay quiet, each will be convicted of the minor offense and spend one year in prison. If one and only one of them finks, she will be freed and used as a witness against the other, who will spend four years in prison. If they both fink, each will spend three years in prison.

Players: The two suspects.

Actions: {Quiet, Fink}.

Preferences:

- Suspect 1's ordering of the action profiles, from best to worst, is (Fink, Quiet) (she finks and suspect 2 remains quiet, so she is freed), (Quiet, Quiet) (she gets one year in prison), (Fink, Fink) (she gets three years in prison), (Quiet, Fink) (she gets four years in prison).
- Suspect 2's ordering is (Quiet, Fink), (Quiet, Quiet), (Fink, Fink), (Fink, Quiet).

Using payoff function we can do this:

- Player1: $u_1(\text{Fink, Quiet}) = 3$, $u_1(\text{Quiet, Quiet}) = 2$, $u_1(\text{Fink, Fink}) = 1$, $u_1(\text{Quiet, Fink}) = 0$.
- Player2: $u_2(\text{Quiet, Fink}) = 3$, $u_2(\text{Quiet, Quiet}) = 2$, $u_2(\text{Fink, Fink}) = 1$, $u_2(\text{Fink, Quiet}) = 0$.

		Suspect 2	
		Quiet	Fink
Suspect 1	Quiet	2,2	0,3
	Fink	3,0	1,1

The Prisoner's Dilemma models a situation in which there are gains from cooperation (each player prefers that both players choose Quiet than they both choose Fink) but each player has an incentive to "free ride" (choose Fink) whatever the other player does.

Pollution game

This game is the extension of Prisoner's Dilemma to the case of many players. Assume that there are n countries and that each country faces the choice of either passing legislation to control pollution or not. Assume that pollution control has a cost of 3 for the country, but each country that pollutes adds 1 to the cost of all countries (in terms of added basic solution concepts and computational issues health costs, etc.). The cost of controlling pollution (which is 3) is considerably larger than the cost of 1 a country pays for being socially irresponsible.

Suppose that k countries choose not to control pollution. Clearly, the cost incurred by each of these countries is k . On the other hand, the cost incurred by the remaining $n - k$ countries is $k + 3$ each, since they have to pay the added cost for their own pollution control. The only stable solution is the one in which no country controls pollution, having a cost of n for each country. In contrast, if they all had controlled pollution, the cost would have been only 3 for each country.

Bach or Stravinsky?

In the Prisoner's Dilemma the main issue is whether or not the players will cooperate (choose Quiet). In the following game the players agree that it is better to cooperate than not to cooperate, but disagree about the best outcome.

EXAMPLE 16.2 (Bach or Stravinsky?) Two people wish to go out together. Two concerts are available: one of music by Bach, and one of music by Stravinsky. One person prefers Bach and the other prefers Stravinsky. If they go to different concerts, each of them is equally unhappy independently of the composer.

Players: The two people.

Actions: {Back, Stravinsky}

Preferences: (Stravinsky, Stravinsky), (Back, Stravinsky), (Stravinsky, Back), (Back, Back)

- Player 1: $u_1(\text{Back}, \text{Back}) = 2$, $u_1(\text{Stravinsky}, \text{Stravinsky}) = 1$, $u_1(\text{Back}, \text{Stravinsky}) = u_1(\text{Stravinsky}, \text{Back}) = 0$
- Player 2: $u_2(\text{Stravinsky}, \text{Stravinsky}) = 2$, $u_2(\text{Back}, \text{Back}) = 1$, $u_2(\text{Back}, \text{Stravinsky}) = u_2(\text{Stravinsky}, \text{Back}) = 0$

		Player 2	
		Back	Stravinsky
Player 1	Back	2,1	0,0
	Stravinsky	0,0	1,2

Matching Pennies

Aspects of both conflict and cooperation are present in both the Prisoner's Dilemma and BoS. The next game is purely conflictual (such a game is called "strictly competitive").

Two people choose, simultaneously, whether to show the Head or the Tail of a coin. If they show the same side, person 2 pays person 1 a dollar; if they show different sides, person 1 pays person 2 a dollar. Each person cares only about the amount of money she receives.

		Player 2	
		Head	Tail
Player 1	Head	1, -1	-1, 1
	Tail	-1, 1	1, -1

Stag Hunt

Each of a group of hunters has two options: remain attentive to the pursuit of a stag, or catch a hare. If all hunters pursue the stag, they catch it and share it equally; if any hunter devotes her energy to catching a hare, the stag escapes, and the hare belongs to the defecting hunter alone. Each hunter prefers a share of the stag to a hare.

Players: The n hunters.

Actions: {Stag, Hare}

Preferences:

- Player _{i} : All Stag $\rightarrow 2$, Player i hare (no matter the others) $\rightarrow 1$, player i stag but at least another hare $\rightarrow 0$

When $n = 2$ we can use the table

		Hunter 2	
		Stag	Hare
Hunter 1	Stag	2, 2	0, 1
	Hare	1, 0	1, 1